SR UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

LAB ASSIGNMENT :- 2 Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI

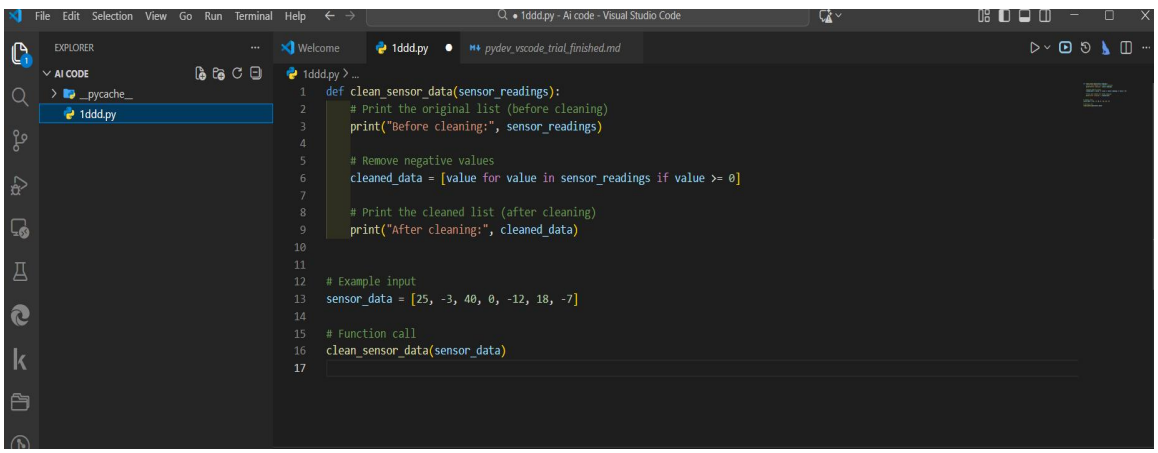
Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name:- Dr. Rishabh Mittal	

STUDENT NAME	PENTAPURIYESHWANTHREDDY
ENROLLMENT NUMBER	2403A51L39
SUBJECT NAME	AI Assisted Coding
COURSE CODE	23CS002PC304
YEAR/SEM	III/II

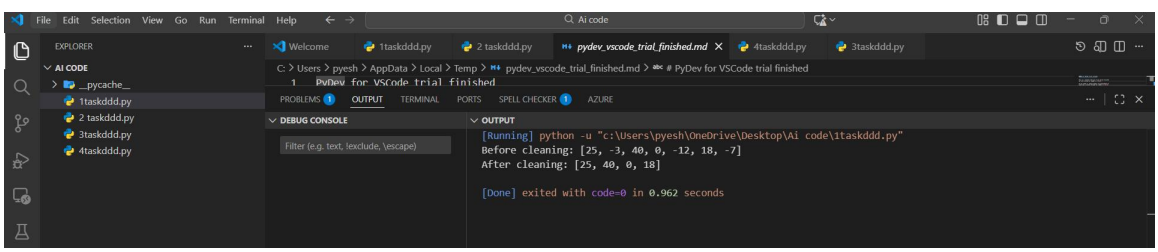
TASK:-1 Cleaning Sensor Data

PROMPT :- You are a Python expert working in Google Colab. Write a Python function that cleans IoT sensor data by removing all negative values from a list of numbers. Show the original list (before cleaning) and the cleaned list (after removing negative values). Include a sample input list and display the output clearly.”

CODE:-



OUTPUT:-



CODE Explanation

1. The Goal
To clean up dirty data. Negative numbers are "dirt"; positive numbers are "clean."

2. The Mental Model: A Sieve
Imagine pouring sand and rocks through a mesh screen.

The Code: The mesh screen.

The Rule: Only "positive numbers" fit through the holes.

The Result: The negative numbers get stuck and are thrown away.

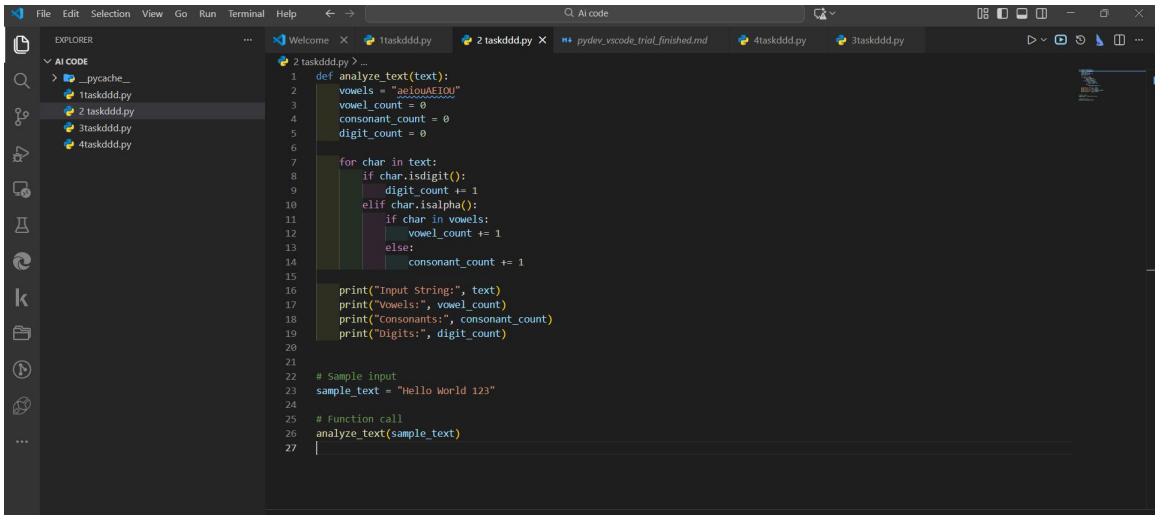
3. Quick Visual
Input: [25, -3, 40] Filter: is number >= 0? Output: [25, 40]

Simple Summary: The code looks at every number and asks, "Are you negative?" If yes, it deletes it. If no, it keeps it.

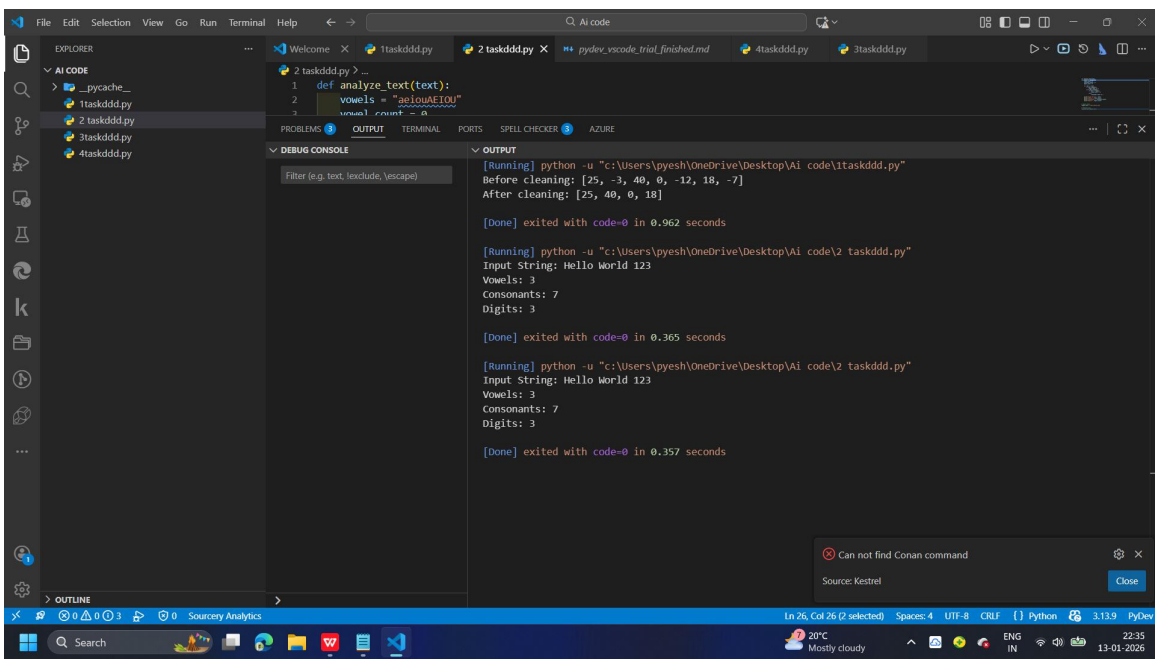
TASK:-2 String Character Analysis

Task 2 Prompt 2:-"Generate a Python function for text analysis that takes a string as input and counts the number of vowels, consonants, and digits in it. The function should clearly distinguish between vowels (a, e, i, o, u – both uppercase and lowercase), consonants (alphabetic characters excluding vowels), and digits (0–9). Ignore spaces and special characters. Display the counts clearly, and include sample input strings along with their expected outputs to demonstrate that the function works correctly

CODE:-



OUT PUT:-



CODE explanation

1.Here is the logic explained in 7 short key points:

2.Setup: The code defines a list of vowels ("aeiouAEIOU") and sets three counters (vowels, consonants, digits) to zero.

3.Looping: It looks at the input text one character at a time (like reading a sentence letter by letter).

4.Digit Check: If the character is a number (0-9), it adds 1 to the digit count.

5.Letter Check: If the character is a letter, it checks if it is inside the "vowels" list.

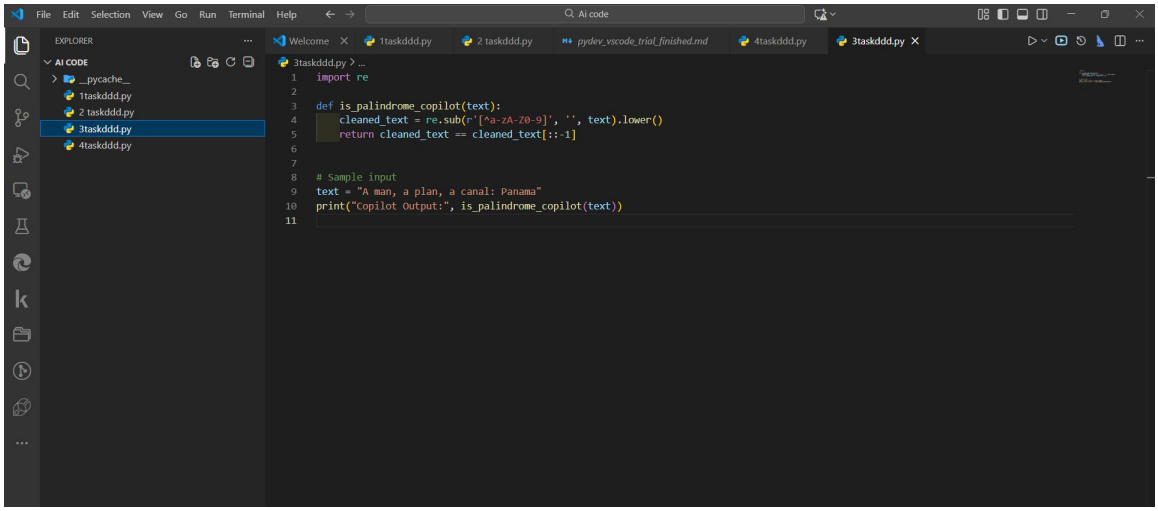
6.Vowel vs. Consonant: If it is in the list, it counts as a vowel; if not, it counts as a consonant.

7.Result: Finally, it prints the original text and the total counts for all three categories.

TASK:-3 Palindrome Check – Tool Comparison

Task 3 prompt 3:-Generate a Python function to check whether a given string is a palindrome. Use Gemini and GitHub Copilot separately to produce the solution. The function should ignore case, spaces, and punctuation. Present the results in a side-by-side code comparison labeled ‘Gemini Version’ and ‘Copilot Version’. After the code, provide brief observations comparing clarity, structure, readability, and efficiency of both implementations, and conclude which version is more maintainable and why.”

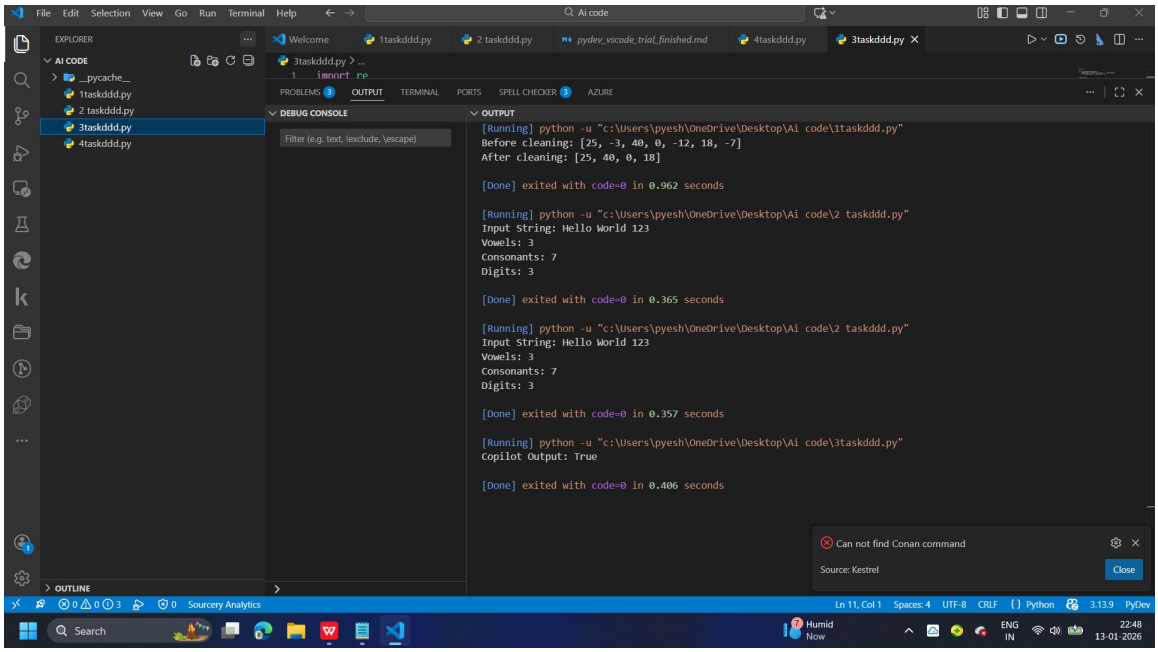
CODE:-


File Edit Selection View Go Run Terminal Help
Ai code
EXPLORER
AI CODE
__pycache__
1taskddd.py
2 taskddd.py
3taskddd.py
4taskddd.py
Welcome
1taskddd.py
2 taskddd.py
pydev_vscode_trial_finished.md
4taskddd.py
3taskddd.py
3taskddd.py > ...
import re

def is_palindrome_copilot(text):
    cleaned_text = re.sub(r'[^a-zA-Z0-9]', '', text).lower()
    return cleaned_text == cleaned_text[::-1]

# Sample input
text = "A man, a plan, a canal: Panama"
print("Copilot Output:", is_palindrome_copilot(text))

OUT PUT:-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Welcome
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2 taskddd.py
pydev_vscode_trial_finished.md
4taskddd.py
3taskddd.py
3taskddd.py > ...
import re
PROBLEMS 3
OUTPUT
TERMINAL
PORTS
SPELL CHECKER 3
AZURE
DEBUG CONSOLE
Filter (e.g. text, !exclude, \escape)
OUTPUT
[Running] python -u "c:\Users\pyesh\OneDrive\Desktop\Ai code\1taskddd.py"
Before cleaning: [25, -3, 40, 0, -12, 18, -7]
After cleaning: [25, 40, 0, 18]
[Done] exited with code=0 in 0.962 seconds
[Running] python -u "c:\Users\pyesh\OneDrive\Desktop\Ai code\2 taskddd.py"
Input String: Hello World 123
Vowels: 3
Consonants: 7
Digits: 3
[Done] exited with code=0 in 0.365 seconds
[Running] python -u "c:\Users\pyesh\OneDrive\Desktop\Ai code\2 taskddd.py"
Input String: Hello World 123
Vowels: 3
Consonants: 7
Digits: 3
[Done] exited with code=0 in 0.357 seconds
[Running] python -u "c:\Users\pyesh\OneDrive\Desktop\Ai code\3taskddd.py"
Copilot Output: True
[Done] exited with code=0 in 0.406 seconds
Can not find Conan command
Source: Kestrel
Close
OUTLINE
Sourcery Analytics
Ln 11, Col 1
Spaces: 4
UTF-8
CRLF
Python
3.13.9
PyDev
Search
Humid
Now
ENG
IN
22:48
13-01-2026

CODE Explanation

1.Here is the logic explained in 6 key points for a solid understanding:

2.Initialize: The code starts with an empty string (cleaned_text) to collect only the characters we care about.

3.Filter Loop: It iterates through every single character in the input string (text).

4.Ignore Junk: The check .isalnum() (Is Alpha-Numeric) filters out spaces, commas, and colons, keeping only letters and numbers.

5.Normalize: It converts every valid character to lowercase using .lower() so that "A" and "a" are treated as the same letter.

6.Reverse Trick: The syntax [::-1] is a Python shortcut that creates a reversed copy of the string.

7.Comparison: It checks if the cleaned_text is exactly equal to its reversed version. If they match, it is a palindrome.

8.Logic Summary: The code effectively turns "A man, a plan..." into "amanaplan...", flips it, and checks if they are identical.

TASK:-4 Code Explanation Using AI

Task 4 prompt 4 :-Explain the following Python function line by line as if teaching a beginner. Choose either a prime-number checking function or a palindrome-checking function. First, display the complete code snippet. Then provide a clear AI explanation for each line, describing what it does and why it is needed. Finally, include student-style comments summarizing the understanding of the code in simple language, mentioning any key concepts learned and possible points of confusion.”

CODE:-

```python
def is_palindrome(text):
    cleaned_text = ""

    for char in text:
        if char.isalnum():
            cleaned_text += char.lower()

    return cleaned_text == cleaned_text[::-1]


# Example usage
text = "Madam, I'm Adam"
print(is_palindrome(text))
```

OUTPUT:-

```
True
PS C:\Users\pyesh\OneDrive\Desktop\Ai code>
```

CODE explanation

Here is the logic in 3 very short points:

Clean: The loop removes all spaces and punctuation, keeping only lowercase letters and numbers (.isalnum() and .lower()).

Flip: The code [::-1] creates a distinct, reversed copy of that clean string.

Match: It compares the two; if the clean string looks exactly the same backwards, it returns True.